

Lecture 12.

Last time: Predicate $P(n), n \in \mathbb{Z}_+$.

$$\begin{array}{c} P(1) \\ \forall k (P(k) \rightarrow P(k+1)) \\ \hline \therefore \forall n \in \mathbb{Z}_+ P(n). \end{array}$$

I Strong Induction

II Recursive Definition

Inference Rule: For $P(n)$ a predicate:

Base Case $P(b)$

"strong" inductive hypothesis

Inductive Step

$$\begin{array}{c} P(b) \\ \forall k \in \mathbb{Z}_+ (P(b) \wedge P(b+1) \dots P(k) \rightarrow P(k+1)) \\ \hline \forall n \geq b (P(n)) \end{array}$$

Proof: Follows from induction by taking
$$Q(n) = Q(b) \wedge Q(b+1) \wedge \dots \wedge Q(n).$$

Theorem: (FTA) If $n \geq 2$, then there are unique
prime numbers $p_1, \dots, p_a$ such that
$$n = p_1 p_2 \dots p_a \quad \text{and} \quad p_1 \leq p_2 \leq \dots \leq p_a.$$

Proof (Existence) We proceed by induction. For $n \geq 2$,
let $P(n) =$ "there exist $a \in \mathbb{Z}_+$, primes $p_1, \dots, p_a$
such that $n = p_1 p_2 \dots p_a$."

Base Case: Observe that $2 = 2$, which is
a prime, so $P(2)$ holds.

Inductive Step : Assume $k \geq 2$ is arbitrary
and $P(2), \dots, P(k)$ hold.
We will show $P(k+1)$.

There are two possibilities:

Case 1 : $k+1$ is prime. In this case,
 $k+1 = k+1$ is a prime factorization
So $P(k+1)$ is true.

Case 2 : $k+1$ is composite. By def.,
we can choose $a, b \geq 2$ such
that $k+1 = ab$. Observe that
 $a < k+1$ and $b < k+1$. By the in-
hyp we can choose $P_1 \dots P_r$ and
 $q_1 \dots q_s$ s.t.

that $a = P_1 \dots P_r$ and $b = Q_1 \dots Q_s$.
But now $n = P_1 \dots P_r Q_1 \dots Q_s$, as desired. $\square$

Remark: "Weak" induction wouldn't suffice here.

Remark: Can think of induction in two ways:

(1) "Bottom up": Given proofs of $P(1) \dots P(k)$,
the inductive step shows how to get a
proof of $P(k+1)$.

(2) "Top down": Given the task of
proving $P(k+1)$, the inductive step
reduces it to the task of proving $P(1) \dots P(k)$.

II Recursive Definitions :

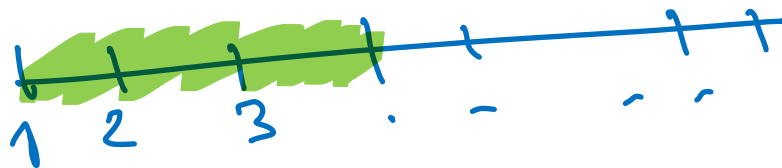
A way to define $f: \mathbb{Z}_+ \rightarrow \mathbb{Z}$ or $f: \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}$

(so far: ① explicit formula eg. $f(n) = n^2$
② using $a_1, a_2, a_3, \dots$ + editing)

New way: to define $f: \mathbb{Z}_+ \rightarrow A$

Base Case: For some $b \geq 1$, define $f(1), \dots, f(b)$

Recursive Step: For every $k \geq b$, define
 $f(k+1)$ in terms of $f(1), \dots, f(k)$.



Ex 1: Fibonacci: $f: \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}$.

Base Case: Define $f(0) = 1, f(1) = 1$.

Recursive Step: For $k \geq 1$, define
 $f(k+1) = f(k) + f(k-1)$.



Key point: each value is defined in terms of
previous ones.

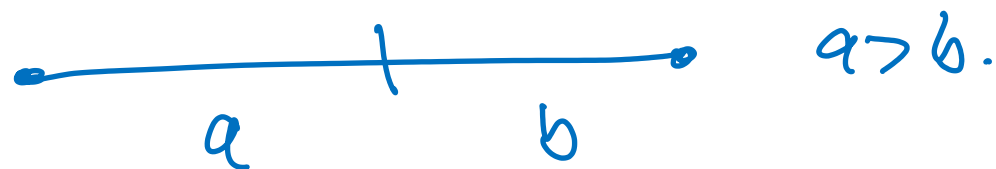
Remark: Recursive defn gives a way to compute
 $f(n)$ by reducing to smaller values of n .

ex2: $f: \mathbb{N}_+ \rightarrow \mathbb{R}$

Base Case: $f(1) = 1.$

Recursive: $f(k+1) = \sqrt{1 + f(k)}.$

Golden Ratio:



Euclid: Consider $\frac{a}{b} = \frac{a+b}{a} = 1 + \frac{b}{a}.$

Let $x = a/b$. ~~A~~ becomes $x = 1 + 1/x.$

$$\equiv x^2 - x - 1 = 0$$

$$\text{So } x = \frac{1 \pm \sqrt{4+1}}{2} = \frac{1 \pm \sqrt{5}}{2} \leadsto$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

Golden Ratio

Amazing fact: $\lim_{n \rightarrow \infty} \frac{f(n+1)}{f(n)} = \phi$.
< show video >.

Theorem: For $n \geq 0$, $f(n) \geq \phi^{n-1}$.

Proof: By induction. Let $P(n) = "f(n) \geq \phi^{n-1}"$.

Base case: $f(0) = 1 \geq \frac{1}{\phi}$ and

$f(1) = 1 \geq \phi^0 = 1$, so

$P(0)$ and $P(1)$ hold.

Inductive Step: Assume $k \geq 1$. Assume $P(1) \dots P(k)$. We will show $P(k+1)$.

Observe that

$$f(k+1) = f(k) + f(k+1)$$

$$\geq \phi^{k-1} + \phi^{k-2}$$

$$= \phi^{k-2} (\phi + 1)$$

by $P(k)$,
 $P(k-1)$

$$= \phi^{k-2} \cdot \phi^2$$

$$= \phi^k, \text{ as desired.}$$

